YuQing Xie

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Education

Massachusetts Institute of Technology **Cambridge, MA** *2022-present*
PhD in EECS
Theoretical Limits of Equivariant Neural Networks (Advisor: Tess Smidt)

Massachusetts Institute of Technology **Cambridge, MA** *2022-2024*
MS in EECS
Equivariant Symmetry breaking Sets (Advisor: Tess Smidt)

Massachusetts Institute of Technology **Cambridge, MA** *2018-2022*
BS in Physics and Math

Selected coursework

- **Physics:** Quantum Field Theory, General Relativity, Statistical Physics, Theory of Solids, Statistical Physics in Biology
- **Math:** Lie Algebras, Algebraic Topology, Algebraic Geometry, Topology, Real Analysis, Complex Analysis, Discrete Math
- **EECS:** Algorithms for Inference, Advanced Complexity Theory, Foundations of Cryptography, Theory of Computation, Shape Analysis

Publications

- **Xie, Y.**, Daigavane, A., Kotak, M., Smidt, T., (2026). "Asymptotically Fast Clebsch-Gordan Tensor Products with Vector Spherical Harmonics". In: *International Conference on Machine Learning*. PMLR. URL: https://openreview.net/forum?id=8UzZ0CbPsk.
- Daigavane, A., **Xie, Y.**, Vani, B. P., Saremi, S., Kleinhenz, J., Smidt, T., (2025). "Matching the Optimal Denoiser in Point Cloud Diffusion with (Improved) Rotational Alignment". In: *arXiv preprint arXiv:2510.03335*. URL: https://doi.org/10.48550/arXiv.2510.03335.
- Zhang, X. et al. (including **YuQing Xie**) (2025). "Artificial intelligence for science in quantum, atomistic, and continuum systems". In: *Foundations and Trends® in Machine Learning* 18.4, pp. 385–912. URL: http://dx.doi.org/10.1561/2200000115.
- **Xie, Y.**, Smidt, T., (2025). "A Tale of Two Symmetries: Exploring the Loss Landscape of Equivariant Models". In: *Advances in Neural Information Processing Systems*. URL: https://openreview.net/forum?id=rH4aGTL4jY.
- **Xie, Y.**, Daigavane, A., Kotak, M., Smidt, T., (2025). "The Price of Freedom: Exploring Expressivity and Runtime Tradeoffs in Equivariant Tensor Products". In: *Forty-second International Conference on Machine Learning*. URL: https://openreview.net/forum?id=EvIwwGYTLc.
- **Xie, Y.**, Smidt, T., (2024). "Equivariant Symmetry Breaking Sets". In: *Transactions on Machine Learning Research*. **J2C certified and presented at ICLR 2025**. URL: https://openreview.net/forum?id=tHKH4DNSR5.
- Hintz, P., **Xie, Y.,** (2022). "Quasinormal modes of small Schwarzschild–de Sitter black holes". In: *Journal of Mathematical Physics* 63.1. URL: https://doi.org/10.1063/5.0062985.
- Hintz, P., **Xie, Y.,** (2021). "Quasinormal modes and dual resonant states on de Sitter space". In: *Physical Review D* 104.6, p. 064037. URL: https://doi.org/10.1103/PhysRevD.104.064037.

Research Experience

Theoretical Limits of Equivariant Neural Networks *07/2022 - present*
Atomic Architects (Prof. Tess Smidt)

- Defined notion of equivariance for spontaneous symmetry breaking systems and completely characterized necessary size of symmetry breaking input for any design based on input and output symmetries. - Published in TMLR 2024
- Clarified tradeoff between expressivity and speed for Clebsch-Gordan tensor product alternatives, explaining massive speed gains claimed in literature. - Published in ICML 2025
- Created first concrete example where equivariance is correct yet hurts optimization. This led to the discovery of a novel mechanism for loss barriers in linearly constrained neural networks. - Published in NeurIPS 2025
- Derived an asymptotic expansion of the matrix Fisher distribution for SO(3). This helps explain rotation alignment in diffusion models. - Preprint
- Derived the asymptotically fastest algorithm for Clebsch-Gordan tensor products based off of a generalization of FFT ideas. - Published in ICML 2026

Experimental Tools for Condensed Matter Physics *06/2021 - 06/2022*
Ju Group (Prof. Long Ju)

- Worked on building the world's third cryogenic atomic force microscope
- Independently built, optimized, and tested a microscope setup for viewing samples such as graphene in a cryogenic dewar

Mathematical Physics of Black Holes *02/2020 - 05/2021*
Prof. Peter Hintz

- Demonstrated convergence of quasinormal modes in Schwarzschild de Sitter spacetime numerically in the limit of small black holes and performed numerical computation of dual resonant states for massless fields in SdS spacetime - Published in Journal of Mathematical Physics 2022
- Resolved the disagreement over existence of QNMs in dS spacetime by computing exact dual resonance modes of massless scalar fields in de Sitter spacetime - Published in Physical Review D 2021

Performance Impacts of Novel Hardware *06/2019 - 09/2019*
Quantum Photonics Group (Prof. Dirk Englund)

- Characterized performance limits of novel energy efficient hardware for general matrix multiplication (GeMM)
- Simulated impact of hardware errors on machine learning performance

Awards

- National Science Foundation Graduate Fellowship (2022)
- MIT Schwarzman College of Computing Fellowship (2022)
- International Theoretical Physics Olympiad - 1st Place Team (2020, 2021)
- William Lowell Putnam Competition - Top 200 (2018,2019)
- International Physics Olympiad (Team USA) - Gold Medal (2018)

Talks

- Asymptotically Fast Clebsch-Gordan Tensor Products with Vector Spherical Harmonics - Simple Persons Applied Math Seminar (SPAMS) (2026)
- Asymptotically Fast Clebsch-Gordan Tensor Products with Vector Spherical Harmonics - Geometry Learning Lab, Northeastern (2026)
- Equivariant Symmetry Breaking - APS March Meeting contributed talk (2024)

Teaching

- Will teach an adapted version of Prof. Tess Smidt's graduate course 6.7970/8.750 during MIT's IAP 2027
- Teacher at MIT Splash 2024 - explained group theory concepts to high schoolers using Rubik's cubes
- Physics Mentorship Program 2022 - provided guidance for undergrad students
- TA for Introduction to Special Relativity 2021 (8.20)

- Physics Instructor for HSSP 2019

Service

- Proceedings chair for Boston TAG party 2026, a joint conference by TAG-DS and Boston Symmetry Day
- Reviewer for ICML, NeurIPS, Nature Machine Intelligence, ICML GRaM workshop, TAG-DS conference
- Volunteer at Cambridge Science Festival 2025 - explained basic AI concepts to families
- Treasurer for House 5 of New House 2021
- Secretary for House 5 of New House 2019

Languages

- **English** [Native]
- **Chinese** [Proficient]

Hobbies

Soccer, Bouldering, Running, A Capella, Guitar, Chess